

Rajalakshmi Engineering College

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2024_28_III_OOPS Using Java Lab

REC_2028_OOPS using Java_Week 4_CY

Attempt : 1
Total Mark : 40
Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

Anjali is preparing a report on text complexity. She wants to identify all words in a sentence that contain at least one digit so she can analyze numeric mentions.

Your task is to write a program that extracts and prints all words containing at least one digit from a given sentence.

If no such word exists, print "No words with digits found".

Input Format

The input contains a single line containing a sentence with multiple words.

Output Format

The output prints all words containing at least one digit separated by a space.

If no word contains a digit, print "No words with digits found".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: The model X100 and Y200 are available

Output: X100 Y200

Answer

```
import java.util.*;
class Main {
    public static void main(String[] args) {
        Scanner in = new Scanner(System.in);
        String s = in.nextLine();
        String[] words = s.split(" ");
        StringBuilder result = new StringBuilder();
        for (String word : words) {
            boolean hasDigit = false;
            for (int i = 0; i < word.length(); i++) {
                if (Character.isDigit(word.charAt(i))) {
                    hasDigit = true;
                    break;
                }
            }
            if (hasDigit) {
                result.append(word).append(" ");
            }
        }
        if (result.length() > 0) {
            System.out.println(result.toString().trim());
        } else {
            System.out.println("No words with digits found");
        }
    }
}
```

Status : Correct

Marks : 10/10

2. Problem Statement

In a university library, librarians need to track the usage of special characters in students' notes.

To help them, you are asked to write a program that counts the number of specific symbols in each passage of text.

The symbols of interest are:

Exclamation marks (!) Colons (:) Semicolons (;)

Input Format

The first line of input contains an integer T, representing the number of test cases (passages).

Each of the next T lines contains a single passage of text.

Output Format

For each test case, print three integers separated by spaces, representing the number of exclamation marks, colons, and semicolons in the passage.

The first line of output corresponds to the first passage, the second line to the second passage, and so on.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 1
Hello! How are you
Output: 1 0 0

Answer

```
import java.util.*;

class Main {
    public static void main(String[] args) {
        Scanner in = new Scanner(System.in);
```

```

int T = in.nextInt();
in.nextLine();

for (int t = 0; t < T; t++) {
    String s = in.nextLine();
    int exclamation = 0, colon = 0, semicolon = 0;

    for (int i = 0; i < s.length(); i++) {
        char c = s.charAt(i);
        if (c == '!') exclamation++;
        else if (c == ':') colon++;
        else if (c == ';') semicolon++;
    }

    System.out.println(exclamation + " " + colon + " " + semicolon);
}
}
}

```

Status : Correct

Marks : 10/10

3. Problem Statement

Meera is practicing her English vocabulary. She wants to focus on words that have more vowels in them, as they help improve her pronunciation. She decides to extract only those words from a sentence that contain at least two vowels.

Your task is to help Meera by writing a program that finds such words from the given sentence.

Input Format

The input contains a string representing the sentence.

Output Format

The output prints all the words that contain at least two vowels, separated by a space.

If no such word exists, print "No words with two vowels".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: This is an example sentence

Output: example sentence

Answer

```
import java.util.*;

class Main {
    public static void main(String[] args) {
        Scanner in = new Scanner(System.in);
        String s = in.nextLine();
        String[] words = s.split(" ");
        StringBuilder result = new StringBuilder();

        for (String word : words) {
            int vowelCount = 0;
            for (int i = 0; i < word.length(); i++) {
                char c = Character.toLowerCase(word.charAt(i));
                if (c == 'a' || c == 'e' || c == 'i' || c == 'o' || c == 'u') {
                    vowelCount++;
                }
            }
            if (vowelCount >= 2) {
                result.append(word).append(" ");
            }
        }

        if (result.length() > 0) {
            System.out.println(result.toString().trim());
        } else {
            System.out.println("No words with two vowels");
        }
    }
}
```

Status : Correct

Marks : 10/10

4. Problem Statement

Neha is analyzing text messages to identify words that have repeated characters. A word is considered "repetitive" if any character appears more than once in that word.

Your task is to write a program that extracts all words that contain repeated characters from a given sentence.

If no such word exists, print "No repetitive words found".

Input Format

The input contains a single line containing a sentence with multiple words.

Output Format

The output prints all words that contain repeated characters separated by a space.

If no word contains repeated characters, print "No repetitive words found".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: letter balloon apple tree

Output: letter balloon apple tree

Answer

```
import java.util.*;

class Main {
    public static void main(String[] args) {
        Scanner in = new Scanner(System.in);
        String s = in.nextLine();
        String[] words = s.split(" ");
        StringBuilder result = new StringBuilder();
        for (String word : words) {
            Set<Character> seen = new HashSet<>();
```

```
boolean isRepetitive = false;
for (int i = 0; i < word.length(); i++) {
    char c = word.charAt(i);
    if (seen.contains(c)) {
        isRepetitive = true;
        break;
    }
    seen.add(c);
}
if (isRepetitive) {
    result.append(word).append(" ");
}
}
if (result.length() > 0) {
    System.out.println(result.toString().trim());
} else {
    System.out.println("No repetitive words found");
}
}
}
```

Status : Correct

Marks : 10/10